

RISHABH VINEETH

Chennai, India | +91 8590246813 | rishabh.vineeth2023@vitstudent.ac.in
linkedin.com/in/rishabh-vineeth-2026b4329 | github.com/Ribash123

Professional Summary

Computer Science and Engineering undergraduate specializing in Artificial Intelligence and Machine Learning with strong foundations in Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, and Computer Networks. Experienced in developing software applications, backend services, RESTful APIs, cloud-based systems, machine learning pipelines, and AI-powered applications using C++, Java, Python, JavaScript, Flask, AWS, and modern machine learning frameworks.

Education

Vellore Institute of Technology (VIT), Chennai <i>B.Tech in Computer Science and Engineering (AI & ML) CGPA: 8.39/10</i>	2023 – 2027 <i>Expected Graduation: June 2027</i>
Private International English School, Abu Dhabi <i>Class XII (CBSE) Percentage: 80%</i>	2022

Technical Skills

Programming Languages: C++, Java, Python, C, JavaScript, SQL

Software Engineering: Data Structures and Algorithms, Object-Oriented Programming (OOP), Software Development, Backend Development, RESTful APIs, API Integration, Debugging, Problem Solving

Core Computer Science: Operating Systems, Database Management Systems (DBMS), Computer Networks, Computer Architecture

AI and Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Transfer Learning, Explainable AI

Frameworks and Libraries: Flask, TensorFlow, Keras, Scikit-learn, Transformers, CodeBERT, Hugging Face

Cloud Technologies: AWS IoT Core, AWS Lambda, Amazon SageMaker

Tools and Platforms: Git, GitHub, Linux, HTML, CSS

Projects

Code Intent Verification and Automated Correction *Python, CodeBERT, Transformers, Scikit-learn, LLMs*

- Engineered a hybrid code-analysis pipeline combining rule-based validation, CodeBERT semantic embeddings, and supervised machine learning to detect inconsistencies between natural-language programming intent and source-code implementation.
- Developed and evaluated a Logistic Regression classification model to categorize code-intent pairs as correct, partially correct, or incorrect using semantic representations of developer intent and source code.
- Integrated an LLM-based automated code correction module to generate code repairs and explanations, achieving 75% classification accuracy and a weighted F1-score of 0.85 on a custom dataset of 60 code-intent pairs.

Finance Web Application – Backend Development and REST APIs *Flask, Python, REST APIs, JavaScript, HTML, CSS, AI*

- Engineered a Flask-based web application for expense tracking, categorization, and financial analysis through an interactive user interface.
- Implemented backend functionality and RESTful API integration to support expense management, data processing, and personalized financial planning.
- Integrated an AI-powered budgeting assistant to analyze user expense patterns and generate personalized spending insights and recommendations.

EnerSense AI – Cloud-Based Energy Monitoring Platform *Python, AWS IoT Core, AWS Lambda, Amazon SageMaker*

- Developed a cloud-based energy monitoring platform integrating 5+ sensors with AWS services for real-time data monitoring, diagnostics, and analytics.
- Built an event-driven processing pipeline using AWS IoT Core, AWS Lambda, and Amazon SageMaker for energy consumption forecasting and anomaly detection.

TourMate – Multilingual NLP and Speech Processing Application *Python, NLP, Speech Recognition, Hugging Face*

- Developed a multilingual travel assistant supporting 3+ languages for location-based tourist assistance and real-time communication.
- Integrated Natural Language Processing and speech recognition components for intent detection, voice-based inter-

action, and multilingual assistance.

Interpretable Brain Tumour Classification with Explainable AI *Python, TensorFlow/Keras, MobileNetV2, Grad-CAM*

- Engineered a deep learning pipeline using MobileNetV2 transfer learning and a custom attention mechanism for brain tumour classification from MRI images.
- Implemented image preprocessing, data augmentation, model fine-tuning, and performance evaluation using TensorFlow and Keras.
- Applied Grad-CAM-based Explainable AI techniques to generate visual explanations highlighting image regions influencing model predictions.

Certifications

- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25)
- NPTEL: Conservation Economics (2024)
- IIT Bombay: C, C++, and Python Programming (2023)

Leadership and Activities

- **Member, ViTeach Club:** Conducted coding workshops and supported peer learning initiatives.
- **Participant, AI Workshop on Agentic AI:** Attended an Ericsson-led workshop focused on agentic AI systems.